# Venkata Lakshmi Parimala Pasupuleti

AI Solutions Architect · LangGraph · RAG Architecture · LLMOps · Eval Pipelines · Responsible AI · Python · AWS · Production AI

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## PROFESSIONAL SUMMARY

Software engineer and AI architect with 8 years shipping production systems and hands-on expertise designing multi-agent orchestration, RAG retrieval strategy, LLMOps pipelines, and responsible AI boundaries. Architectural decisions I own: **Leasel** - a 3-surface platform sharing one evidence model, where I chose the surface boundaries, designed the 8-agent StateGraph, made the build-vs-buy call on RAG (hand-rolled TF-IDF + semantic reranking over LangChain - controllability over convenience), and defined where AI authority ends and human authority begins in every regulated decision. **AgentProbe** - defined LLMOps standards from scratch: eval pipeline architecture, prompt versioning, CI regression harness with golden datasets, LLM-as-judge scoring, and offline + online eval loops. The open-source result is what I'd put in front of enterprise clients as a reference architecture. At Entrata (8yr): designed and maintained data pipelines in a regulated domain (HUD/PHI compliance) where architectural decisions had direct compliance consequences - the discipline that transfers directly to responsible AI governance. Python, TypeScript, AWS (S3/Bedrock/Lambda), LangGraph, Docker, Kubernetes. Active Claude + Cursor user. Case study: github.com/venkatalakshmiparimala/AgentProbe

## CORE COMPETENCIES

AI Solutions Architecture | Multi-Agent Orchestration | LangGraph / LangChain | Amazon Bedrock | RAG Architecture & Hybrid Retrieval | Reranking & Embedding Selection | Context Engineering | LLMOps / MLOps | Eval Pipelines & LLM-as-Judge | Prompt Versioning & CI/CD | Offline + Online Evaluation | Guardrails & Grounding Controls | Responsible AI & Governance | Prompt-Injection Defense | PII & Data Isolation | HITL / Human-in-the-Loop | Fine-Tuning vs RAG vs Prompting | Tool Use / Tool-Calling | Chunking & Document Ingestion | Agent Tracing & Observability | Event-Driven Systems | Streaming Inference | AWS (S3 / Bedrock / Lambda) | Cloud-Native Architecture | Microservices / Serverless | Docker / Kubernetes | Python / TypeScript / FastAPI | Technical Mentoring

## EXPERIENCE

### Blue Horse Digital

Sep 2026 - Present

**AI/ML Engineer (Volunteer) — Virginia, USA**

- Building ML models and AI-powered data pipelines; integrating AI components with backend/frontend teams; surfacing business insights into application-facing APIs.

### Magizhs Tech Solution

May 2025 - Aug 2025

**AI/ML Engineer Intern**

- Designed and built an **agentic mission-planning system** as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing); implemented an MCP server for downstream tool integration; RAG layer over FAA Part 107 corpus with deterministic guardrails blocking non-compliant missions before export.

### George Mason University

Aug 2024 - Aug 2025

**Research Associate / Graduate Teaching Assistant — Virginia, USA**

- Fine-tuned a custom classification model (HuggingFace Transformers + PyTorch); deployed inference API on Quart; made explicit build-vs-buy decision - fine-tuning over prompting for fixed-schema classification task with labeled training data available.
- Built multi-source prediction models (Python, R) on 190+ country datasets ($R^2$=0.977); applied Isolation Forest anomaly detection across 3 research pipelines; mentored 600+ students across MIS 303 sections.

### Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

**Senior Software Engineer — Lehi, UT (Remote)**

- **Owned regulated data pipeline architecture for 7 years** - HUD income-limit ingestion (few thousand rows → ~180,000 domain records powering eligibility, rent, and HAP calculations); designed validation, reconciliation, and rollback logic for every annual cycle; **zero compliance violations** - the same responsible-AI governance discipline applied to data pipelines.
- **Designed and maintained AWS S3 migration with 100% PHI compliance** - Libsodium-encrypted pipelines, multi-tenant data isolation at the schema layer, zero security incidents; HIPAA-compliant architecture for 100K+ monthly active users.
- **Architectural decisions on microservices redesign** - led Docker containerization (50% scalability improvement, 30% deployment time reduction); optimized 200+ SQL queries (30% performance gain) across complex multi-tenant financial schemas.
- Shipped 20+ full-stack production modules across 120+ Agile sprints; eliminated 80% of manual workload via automation pipelines; mentored 5+ junior engineers; conducted 300+ peer code reviews.

### Entrata India - Db Xento Systems

Jan 2017 - Sep 2018

**Associate / Trainee Software Engineer — Pune, India**

- Built property management and compliance modules (PHP, Python, Java, PostgreSQL); led AWS S3 cloud migration of client documents; income limits import (1.6M HUD/Tax Credit records across 5 years).

## PROJECTS

### Leasel: Multi-Agent AI Architecture - 3 Surfaces, One Evidence Model

LangGraph · RAG Architecture · Responsible AI · LLMOps

Architecture decision record: 3 user surfaces (support engineers, tenants, staff) sharing one SQLite knowledge repository - designed the surface boundary rules, safety profiles, and data isolation layer. **8-agent LangGraph StateGraph** with durable state, conditional routing, and **tool-calling** between agents; **HITL escalation gate** when confidence drops below threshold; **build-vs-buy on RAG** - hand-rolled TF-IDF + semantic reranking over LangChain with **section-aware chunking** for multi-section regulatory PDFs; **agent tracing** and observability at every node transition; deterministic compliance boundary enforced in code: **LLM extracts and reasons; deterministic logic makes every regulated call**. Responsible AI: LLM never auto-decides eligibility - safety boundary tested and documented. Eval: 93.6% precision, 87.9% F1 on 1,240 test cases; CI regression harness gates merges.

Python · LangGraph (8-agent) · FastAPI · TF-IDF · Semantic Reranking · OCR · SQLite · 2026

### AgentProbe: LLMOps Reference Architecture + Eval Toolkit

LLMOps Standards · Eval Pipelines · Prompt Versioning · CI/CD

Defined LLMOps standards from scratch: eval pipeline architecture, prompt versioning across providers, CI-integrated regression harness with golden datasets, **LLM-as-judge scoring**, **offline + online evaluation loops**, latency and hallucination rate monitoring, **agent tracing** at every inference step. 15-endpoint FastAPI backend; model routing across OpenAI/Anthropic/Groq/Ollama; zero vendor dependency; **streaming inference** patterns; **tool-calling** and structured output validation across all providers. Used as the evaluation infrastructure for Leasel (1,240+ test cases). Live reference architecture: github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · OpenAI / Anthropic / Groq / Ollama · 2025

### AI on FHIR: NLP-Driven Medical Records Query System

FHIR · HIPAA · Responsible AI · Clinical NLP

Built a Flask + React NLP platform translating natural-language queries (EN/ES) into **FHIR API calls** using LLMs, BERT, and embeddings. Enforced **HIPAA-compliant RBAC** at the data layer - PHI access controlled per role, not per request. Reduced manual clinical retrieval time ~60%. Responsible AI design: PII and PHI never passed to LLM; access control enforced architecturally, not via prompt.

Flask · React · FHIR APIs · HIPAA RBAC · LLMs · BERT · spaCy · Embeddings · Docker · 2024

## EDUCATION

George Mason University **- M.S. Data Analytics Engineering** May 2026

GPA: 3.87 · Virginia, USA · Coursework: ML systems, NLP, cloud infrastructure, statistical modeling, deep learning

Jawaharlal Nehru Technological University **- B.Tech Computer Science & Engineering** 2017